

Results

gen0: imitating the teacher (2026-09-26)

Question: can the current small network (63k parameters) learn every act-1 fight, or does it need to be bigger or different?

Answer: it can. One net trained on all fights plays almost as well as the teacher everywhere. On Slime Boss it plays exactly as well as the net trained on Slime Boss only. No architecture change is needed yet.

	gen0 vs teacher
All 4,810 dev fights	-1.2 HP-eq per fight (95% CI -1.5 to -0.8); wins 93.0% → 91.6%
Over a whole act-1 run	about -6.5 HP-eq , mostly from the boss (-2.4) and the elites (-1.3)
Slime Boss	-2.9 HP-eq; wins 74.7% → 69.9%
Elites	lagavulin -2.2, sentries -1.9, gremlin nob +1.0
Hallway fights	-0.2 to -1.3 HP-eq each; almost no extra deaths
Slime Boss, vs the slime-only net	-0.1 HP-eq (CI -1.6 to +1.4); both won 698 of 999

Where the gap is: in lost fights, not in HP. On fights both players win, HP lost is within about 0.7 HP everywhere. The net loses a few more boss and elite fights than the teacher. That matches the slime-only net, which self-play then fixed.

Training: 1.82M positions, 12 epochs, 45 min on one CPU thread. Nearly all of the learning happened in the first epoch, and there was no overfitting. The run is `value_net_v1/2026-09-26/act1-gen0`.

Runs: `combat_v3/2026-09-26/act1-teacher-dev` (teacher, 35 min) and `combat_v3/2026-09-26/act1-gen0-dev` (gen0, 76 min, before the search speed-up). The confirm set is untouched.

Per-encounter table: `results/gen0-vs-teacher.md`.

gen1: one round of self-play (2026-09-26)

Question: does one round of self-play fix the boss and elite losses that gen0 had, as it did on Slime Boss alone?

Answer: yes on the boss, only partly on the elites. gen1 now beats the teacher overall, and gen0 by a wide margin. Almost all of the gain over the teacher is on Slime Boss. Among the elites, Lagavulin is fixed, but Three Sentries is still slightly behind the teacher.

	gen0 vs teacher	gen1 vs teacher	gen1 vs gen0
All 4,810 dev fights, HP-eq per fight	-1.2 (-1.5, -0.8)	+0.6 (+0.3, +0.9)	+1.8 (+1.4, +2.1)
Wins	93.0% → 91.6%	93.0% → 93.5%	91.6% → 93.5%
Over a whole act-1 run, HP-eq	-6.5 (-8.3, -4.7)	+2.4 (+0.7, +4.1)	+8.9 (+7.2, +10.7)

Values in brackets are 95% CIs. Positive means the first player named is better.

Where the difference comes from (HP-eq per act-1 run)

Each encounter's result is weighted by how often a run meets that encounter.

part of the run	gen0 vs teacher	gen1 vs teacher	gen1 vs gen0
Boss (Slime Boss)	-2.4	+2.7	+5.1
Elites	-1.3	-0.4	+0.9
Hallway fights	-2.9	+0.1	+3.0
Events	0.0	0.0	0.0
Total	-6.5	+2.4	+8.9

 **Chart placeholder:** stacked or grouped bars of HP-eq per run by part (boss / elites / hallway), one group each for gen0 vs teacher and gen1 vs teacher. Shows the boss contribution going from negative to positive.

Boss and elites

encounter	fights	win %: teacher / gen0 / gen1	gen1 vs teacher, HP-eq per fight	gen1 vs gen0, HP-eq per fight
Slime Boss	999	74.7 / 69.9 / 78.3	+3.3 (+1.9, +4.7)	+6.2 (+4.8, +7.6)
Gremlin Nob	250	94.8 / 96.4 / 95.2	+0.6 (-0.6, +1.7)	-0.4 (-1.4, +0.5)
Lagavulin	250	92.4 / 88.4 / 91.6	-0.1 (-1.8, +1.6)	+2.1 (+0.5, +3.7)
Three Sentries	250	90.4 / 87.2 / 87.2	-1.5 (-3.2, +0.2)	+0.4 (-1.1, +1.9)

- **Slime Boss:** gen1 wins 93 fights the teacher loses, and loses 57 that the teacher wins (McNemar $p = 0.004$). On fights both win, it also keeps more HP: it loses 18.3 HP per fight against the teacher's 20.4.
- **Lagavulin:** back to the teacher's level. gen0's extra losses (1 fight won only by gen0, 11 won only by the teacher) are gone (6 / 8).
- **Three Sentries:** the one gap left. It has the same win rate as gen0 and loses 14 fights the teacher wins, against 6 the other way ($p = 0.12$). This is the natural target for gen2.
- **Gremlin Nob:** no real difference among the three players.

 **Chart placeholder:** grouped bars of win % for teacher / gen0 / gen1 on Slime Boss and the three elites.

 **Chart placeholder:** forest plot of gen1 vs teacher HP-eq per fight with 95% CIs, one row per encounter (boss and elites at the top). Shows at a glance that only Slime Boss is clearly different.

Hallway fights and events

- **Hallway fights:** gen1 matches the teacher on every encounter, within -0.6 to +0.8 HP-eq per fight, with at most 2 extra deaths per encounter. It is clearly better than gen0 on most of them, by about +0.5 to +2 HP-eq, because it loses less HP.
- **Events:** each one has only 25–89 fights, so the results are noisy. event/lagavulin_event is -2.9 against gen0 (31 fights, one extra loss), which is not a reliable signal.

How gen1 was made

1. **Self-play:** gen0 replayed 4,076 train fights (`act1-a20-8`, `run_seed % 10 = 4`) with new randomness. It won 3,787 of them. Run: `combat_v3/2026-09-26/act1-selfplay-gen1` (19 min).
2. **Fine-tune:** training started from gen0's weights on 52.5k positions from those fights, labelled with how each fight actually ended. Validation error was lowest after epoch 1, and that is the checkpoint kept. Later epochs overfit slightly (validation MSE 0.0071 → 0.0075). Run: `value_net_v1/2026-09-26/act1-gen1` (1 min).
3. **Evaluation:** the same 4,810 dev fights and search settings as gen0. Run: `combat_v3/2026-09-26/act1-gen1-dev` (31 min).

The confirm set is untouched.

Caveats: this is one round of self-play, evaluated on dev only. The differences on elites are within noise at 250 fights per encounter. The best checkpoint came from epoch 1, which suggests the self-play set is small for 10 epochs. More self-play fights, or fewer epochs, are worth trying in gen2.

Per-encounter tables: `results/gen1-vs-teacher.md`, `results/gen1-vs-gen0.md`.